Online Supplement to "Survey of Uncertainty Quantification in Large Language Models: Detection, Mitigation, and Human-Centered and Agentic Perspectives"

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This document is the online supplement to the survey article of the same title. It is not part of the article. It contains the per-paper literature matrix (Table S1), the per-paper coding table (Table S2), the update-search log (Table S3), the disposition of the 126 distinct records returned by that search (Table S4), and the cost and deployment table by method family (Table S5). Section and table numbers cited as "of the article" refer to the article.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **General and reference** → **Surveys and overviews**;

Additional Key Words and Phrases: uncertainty quantification, large language models, hallucination detection, calibration, conformal prediction, human-centered uncertainty quantification

ACM Reference format:
Iqra Fareed and Michael W. Totaro. 2026. Online Supplement to "Survey of Uncertainty Quantification in Large Language Models: Detection, Mitigation, and Human-Centered and Agentic Perspectives". *ACM Comput. Surv.* 0, 0, Article 0 (January 2026), 25 pages.
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Table S1. Literature Matrix: Uncertainty Quantification in Large Language Models – Detection, Mitigation, and Human-Centered UQ.

#	Paper	Problem	UQ Technique	Key Results	Limitations
1	Language Models (Mostly) Know What They Know (Kadavath et al.)	Do LLMs know when their own answers are likely correct?	Self-verbalized (P(True); P(IK) is a trained value head, not verbalized text)	Large models well-calibrated on MC/T-F tasks with suitable formatting; P(True) improves with scale and with showing several own samples; RLHF policies appear miscalibrated, largely fixed by a temperature adjustment	P(IK) needs fine-tuning with a value head (white-box); P(IK) calibration degrades out of distribution; mainly pretrained models and five sampling datasets

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
2	SelfCheck-GPT (Manakul, Liusie, Gales)	Detect hallucinations without external knowledge base or logit access	Consistency-based (semantic-similarity)	Prompt variant best (AUC-PR 93.42 NonFact); NLI close second (92.50), better cost/performance trade-off; beats grey-box token-probability baselines	Computationally heavy (Prompt variant); requires many samples (20); one dataset (GPT-3 WikiBio); can't catch consistent-but-wrong hallucinations (see SAC3)
3	Semantic Uncertainty (Kuhn, Gal, Farquhar)	Token-level entropy overestimates uncertainty when many phrasings share one meaning	Semantic-similarity (Semantic Entropy)	Beats predictive entropy, length-norm entropy, lexical similarity, P(True); gap widens with model size; temp ˜0.5 optimal	Hard clustering (the NLI-based equivalence is not strictly transitive, which Kuhn et al. acknowledge but report as not a noticeable problem in practice; SAR's authors judged ˜36.7% of entailment predictions undesirable in a manual check of 120 long-answer questions); needs logits; ignores token relevance

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
4	Survey of Uncertainty Estimation in LLMs: Sources, Methods, Applications, Challenges (He et al.)	No prior survey organizes UQ by SOURCE (data, model, user) rather than method	All (organizes token-level, Bayesian, consistency-based, conformal, single/multi-prompt, ensembles)	Introduces data/model/user uncertainty taxonomy; 19 open challenges incl. lack of ground-truth uncertainty scores, human meta-evaluation gap	No original empirical validation; long and broad in scope; some overlap with Shorinwa taxonomy
5	SELAUR (Zhang, Liu, Cheng, Wang, Murray, Wei)	Standard RL discards failed agent trajectories entirely, wasting information	Token-level (entropy, LC, margin) used as an RL reward signal, not just detection	Highest scores vs PPO, RLOO, GRPO, GiGPO on both benchmarks with Qwen2.5-1.5B/7B, but small margins over GiGPO in several settings; combined uncertainty beats any single metric; maintains higher entropy late in training (less exploration collapse)	Only tested on 2 benchmarks; requires white-box logit access; adds RL training complexity
6	Uncertainty Quantification in LLM Agents: Foundations, Emerging Challenges, Opportunities (Oh et al.)	Existing UQ assumes static single-turn inference; doesn't fit multi-turn interactive agents	General framework subsuming token-level, semantic, verbalized methods as special cases	Token-probability and verbalized estimators show close-to-random task-success prediction in many cases (tau2-bench, GPT-4.1/Kimi-K2.5); 4 challenges identified with preliminary numerical evidence	Framework is descriptive/position-oriented, not yet a working solution; pilot experiments are small-scale

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
7	A Survey of Confidence Estimation and Calibration in LLMs (Geng et al.)	No comprehensive survey systematically organizing LLM confidence/ calibration work	All (logit-based, internal-state, linguistic, consistency-based, surrogate models)	Organizes by generation vs. classification tasks; white-box/ black-box Venn-diagram taxonomy; explicitly notes lack of experimental benchmarks as own limitation	Self-acknowledged: no original experiments; narrower scope than Shorinwa/He surveys
8	How Humans and LLMs Differ in Processing Uncertainty in Polarized Discourse (Donkers, Ziegler)	Do LLMs perceive/ process expressed uncertainty in discourse the way humans do?	Not a detection method — studies communicative/ social uncertainty perception	Both humans & LLMs detect polarized vs. moderate discourse, but LLMs amplify differences 3-6.5x; humans show cascading mediation, LLMs show independent parallel processing with no cross-construct mediation	Single topic (UBI) tested; LLM-generated (not human) discourse; correlational (structural-equation) evidence, so the "architecture" reading is ours to caveat, not a causal result
9	A Survey on Uncertainty Quantification of LLMs: Taxonomy, Open Research Challenges (Shorinwa et al.)	Field lacks a taxonomy unifying disparate UQ methods considering LLM-specific architecture	4-category taxonomy: token-level, self-verbalized, semantic-similarity, mechanistic interpretability	Most comprehensive method-based taxonomy of the 3 surveys; unique mechanistic-interpretability coverage; robotics + chatbot applications; 'consistency is not factuality' / 'entropy is not factuality' challenges named explicitly	No original experiments; very long; mechanistic-interpretability-UQ connection still speculative

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
10	Language Models Prefer What They Know (Shrivastava, Kumar, Liang)	Absolute self-rated confidence (0-1) is coarse-grained and poorly discriminative	Self-verbalized (relative confidence)	Beats absolute/ self-consistency prompting for 4/5 models (gains about 1–6 points over direct); best rank aggregator varies by model; CoT slightly lowers AUC (87.2 to 86.8, GPT-4o)	Only tested on MC QA; Claude 3.5 Sonnet is an exception; relies on many pairwise LLM calls
11	SAC3 (Zhang, Li, Das, Malin, Kumar)	Self-consistency alone misses question-level and model-level hallucinations	Consistency-based (extends SelfCheck-GPT)	On all-hallucinated classification data, SAC3-all (Guanaco-33b verifier) beats the SC2 self-consistency baseline by 51.2 accuracy points (99.4 vs 48.2, prime numbers); SAC3-Q alone also far exceeds SC2; SAC3-Q and SAC3-all most robust to the threshold	Verifier LM choice matters a lot (verifiers refused many senator questions); extra cost from perturbation, equivalence checks and verifier calls; one target model (gpt-3.5-turbo), four datasets
12	Kernel Language Entropy (Nikitin, Kossen, Gal, Marttinen)	Semantic Entropy's hard clustering discards fine-grained similarity information	Semantic-similarity (generalizes Semantic Entropy; works without token probabilities)	Heat-kernel variant beats SE in 72% of 60 model-dataset scenarios by AUROC (discrete SE 77%, P(True) 83%); SE is a special case of KLE	Needs multiple samples and an NLI model; kernel hyper-parameters; QA and math only; token-likelihood methods do better on non-instruction-tuned models

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
13	Can LLMs Express Their Uncertainty? (Xiong et al.)	No systematic empirical framework compares black-box confidence-elicitation strategies	Self-verbalized + consistency (unifying framework)	Plain verbalized confidence is overconfident (mostly 80–100%) and near chance at failure prediction; multi-sample consistency raises AUROC (GSM8K, GPT-3.5: 54.8 to 92.7) but unevenly, and adding verbalized confidence to the aggregation beats consistency alone	CoT lowers ECE largely via accuracy gains, while AUROC falls on 2 of 5 datasets (GPT-3.5); no strategy consistently best; hard on professional knowledge; single-answer QA only
14	Shifting Attention to Relevance / SAR (Duan et al.)	Entropy-based UQ treats all tokens/ sentences equally, letting irrelevant tokens dominate uncertainty score	Relevance-weighted uncertainty	Higher AUROC than PE, LN-PE, Semantic Entropy and Lexical Similarity in most settings (+7.1 AUROC points over SE on average, four instruction-tuned models, Rouge-L 0.5); SentSAR with 2 samples beat 5-sample baselines on SciQ/TriviaQA	Requires token logits; adds sentence-similarity overhead; not better everywhere (e.g., Rouge-L 0.3, LLaMA-13b TriviaQA; MedQA with Vicuna-13b); correctness defined by Rouge-L/ similarity thresholds; free-form QA, models $\leq$33B

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
15	Benchmarking UQ Methods with LM-Polygraph (Vashurin et al.)	Evaluation protocols for LLM UQ differ across papers (arbitrary correctness thresholds, n-gram quality metrics, omitted simple baselines), making results hard to compare	All categories (Information-based, Sample Diversity, Density-based, Reflexive)	MSP strong on short outputs but lags on GSM8k and selective generation; NLI-based black-box sample-diversity methods best on long outputs; CCP best on claim-level fact-checking in 4 of 5 settings; P(True) mostly no better than random on 7B/12B base models	Models are ≤13B plus GPT-4o-mini; fact-checking labels come from GPT-4 (F1 0.80–0.93 vs. manual on small subsets); main text compares only six methods across languages
16	Position: UQ Needs Reassessment for LLM Agents (Kirchhof, Kasneci, Kasneci)	Definitions of aleatoric and epistemic uncertainty conflict and, the authors argue, lose their meaning when an agent interacts with a user	Position paper: review of aleatoric/ epistemic definitions (mostly classical ML literature) and a research agenda	Toy example shows definitions disagree; proposes three research directions (underspecification, interactive learning, output uncertainties) while keeping aleatoric/ epistemic for training and active learning	No new method or experiments; interactive-agent argument is conceptual, not tested on agents

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
17	Fact-Checking via Token-Level UQ / CCP (Fadeeva et al.)	Token-probability methods conflate claim-type, surface-form, and factual-claim uncertainty	Token-level (Claim-Conditioned Probability)	Highest ROC-AUC vs. Max Probability, Perplexity, Token Entropy, P(True) in all 4 English models (+0.04 to +0.07) and most others, but margins of only 0.01 for Yi and GPT-4 (Arabic); on 100 human-labelled Vicuna-13b claims, ROC-AUC 0.78 vs. FactScore 0.72	P(True) has higher PR-AUC on GPT-3.5-turbo (authors suggest the API's 5-alternative limit); claim splitting uses GPT models; NLI classifier untested across domains; white-box only; calibration not studied
18	Generating with Confidence (Lin, Trivedi, Sun)	Uncertainty (input-level) and confidence (response-specific) are conflated in prior black-box work	Response-similarity graph (black-box)	NLI-entailment similarity usually best (Jaccard/ contradiction variants 0–6 AUARC points lower); response-specific confidence usually beats input-level uncertainty; matches or beats SE and beats P(True) in most settings without logits, but SE wins e.g. LLaMA2/TriviaQA	Degenerate at temperature 0 (stated by authors); SE better at temperature 0.5 with few samples; calibration set aside (appendix: histogram binning works); QA only; correctness judged by gpt-3.5-turbo; models $\leq$13B plus gpt-3.5-turbo

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
19	Detecting Hallucinations in LLM Generation: A Token Probability Approach (Quevedo et al.)	Can four token-probability features from an evaluator LLM, fed to a small supervised classifier, detect hallucination?	Token-probability features with a supervised classifier (evaluator may differ from generator)	HaluEval, trained on 10% and tested on 90%: up to 0.98 (summarization) and 0.95 (QA) accuracy; baselines not rerun on the same split, and about 0.99 even without the condition text, which the authors read as a dataset pattern; trails MIND on 5 of 6 HELM generators	Roughly chance on True-False; collapsed on HaluEval general queries; needs labelled data and token probabilities from an evaluator model
20	Mitigating LLM Hallucinations via Conformal Abstention (Yadkori et al.)	Consistency-based uncertainty scores lack a principled threshold for deciding when to abstain	Conformal abstention (static threshold)	Match-count scores abstain far less than log-probability scores on long answers (Temporal Sequences), comparable on short (TriviaQA); calibrated methods are mostly, not always, under target at small calibration sizes, with very high abstention	Needs a held-out calibration set and exchangeability; one model (Gemini Pro), two datasets; guarantee is marginal and in expectation for the CRC variant; extra sampling cost

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
21	Learning Conformal Abstention Policies (Tayebati et al.)	Static conformal-prediction thresholds don't adapt to varying task complexity/ distribution shift	Conformal prediction (learned/ adaptive, task-level)	Learned policy consistently outperforms fixed-threshold conformal baselines while meeting the same coverage target	Authors caution the learned policy may overfit or distort the conformal guarantee
22	Hallucination Detection and Mitigation in Large Language Models (Pesaranghader, Li)	Industry needs an operational, end-to-end framework connecting detection to mitigation, not just a new metric	Multi-method operational framework	Describes a closed feedback loop across model/ context/data tiers; financial extraction case study lists expected benefits only	No measured results at all; single case study domain (finance); own bibliography contains miscited references
23	Adaptive Conformal Prediction for Improving Factuality (Rubashevskii et al.)	Existing conformal-prediction factuality methods aren't prompt-adaptive, causing over- or under-coverage per prompt	Conformal prediction (prompt-adaptive)	Outperforms global-threshold conformal factuality on category-wise conditional coverage while retaining marginal coverage	Very recent preprint, under review; white-box models only; theorem stated for oracle quantile

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
24	From Calibration to Collaboration (Devic et al.)	UQ research optimizes calibration metrics divorced from whether uncertainty actually helps human decision-making	Meta-level critique of evaluation methodology across all UQ categories	Argues good calibration ≠ good human-AI collaboration; identifies 3 systematic blind spots in current benchmarks/ metrics	No new experiments
25	Teaching Models to Express Their Uncertainty in Words (Lin, Hilton, Evans)	Token log-probability conflates uncertainty over wording with uncertainty over the underlying claim	Self-verbalized (verbalized probability)	GPT-3 can be fine-tuned to state calibrated confidence in words or numbers, moderately calibrated under shift; beats logit baselines on the multi-answer set only; tentative evidence of latent pre-trained features	Arithmetic only, one model (GPT-3 175B); logit methods better on multiply-divide; much weaker out of distribution; supervised fine-tuning
26	The Internal State of an LLM Knows When It's Lying (Azaria, Mitchell)	Can hidden-layer activations reveal truthfulness independent of the model's own output?	Mechanistic / internal-state probing	About 71% (OPT-6.7B) to 83% (LLaMA-2-7B) on held-out topics of a true/false dataset; weaker (about 61–71%) on the model's own generated statements; beats few-shot and BERT baselines (about 54–56%)	Needs labeled training data and white-box access; best layer varies by model; only two 7B models, English, isolated statements

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
27	Discovering Latent Knowledge in LMs Without Supervision (Burns, Ye, Klein, Steinhardt)	Can a truth direction be recovered from activations with zero labels at all?	Mechanistic / internal-state probing (unsupervised)	71.2% vs 67.2% calibrated zero-shot (average excluding T0); about half the prompt sensitivity; robust to misleading prompts	Needs white-box access; about 12 points below the supervised ceiling (83.7%); worse than zero-shot on RoBERTa; yes-no tasks only; deception untested
28	Detecting Hallucinations in Large Language Models Using Semantic Entropy (Farquhar, Kossen, Kuhn, Gal)	Can entropy over meanings detect confabulations without task-specific data, including in paragraph-length text?	Semantic-similarity (semantic entropy, plus a discrete variant needing no token probabilities)	Mean AUROC 0.790 over 30 task-model combinations vs 0.687–0.698 for naive entropy, P(True) and embedding regression; discrete variant beats baselines on GPT-4 biographies (150 claims, 21 people)	Needs about ten samples and an entailment judge; does not catch consistent errors; clustering can over-split; no clinical or legal evaluation
29	Semantic Entropy Probes (Kossen, Han, Razzak, Schut, Malik, Gal)	Can semantic entropy be approximated cheaply, without resampling multiple full generations?	Mechanistic approximation of semantic-similarity signal	Near-zero test-time overhead versus 5–10 samples; generalizes better to unseen tasks than correctness-trained probes; in-distribution similar to them; below full Semantic Entropy	Approximation that does not match full SE; evaluated on QA with models up to 70B; sits ambiguously between semantic and mechanistic categories

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
30	Self-Contradictory Hallucinations of LLMs (Mündler, He, Jenko, Vechev)	Can a model's self-contradiction within one generation be detected, and removed, by prompting alone?	Consistency-based (within-generation)	15.7–22.9% of sentences self-contradict across four models (17.7% ChatGPT); detection F1 about 80%; revision removes 76–90% of contradictions	Roughly 80–90 times the generation tokens; weak with open-source detectors; open-domain text and short QA only
31	Benchmarking LLMs via Uncertainty Quantification (Ye, Yang, Pang, Wang, Wong, Yilmaz, Shi, Tu)	Can conformal prediction set size rank LLMs on uncertainty as well as accuracy?	Conformal prediction used as a benchmark metric (compares models, not UQ methods)	Across 9 model series and 5 multiple-choice tasks: higher accuracy does not imply lower uncertainty, larger models can be more uncertain, instruction fine-tuning tends to raise uncertainty; coverage near the 90% target	Multiple-choice only (free-form loses the guarantee); needs option logits or about 50 samples; calibration and test data must match
32	Inference-Time Intervention (Li, Patel, Viégas, Pfister, Wattenberg)	Can a model be steered toward truthfulness at inference time without fine-tuning?	Mechanistic intervention	Raises Alpaca truthfulness on TruthfulQA from 32.5% to 65.1%; needs only a few hundred labeled examples; no weight fine-tuning	Needs white-box access and labeled data; truthfulness–helpfulness trade-off; mainly TruthfulQA and 7B models, small gains elsewhere

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
33	Robots That Ask for Help / KnowNo (Ren, Dixit, Bodrova, Singh, Tu, et al.)	When should a robot ask a human for help, given uncertainty in LLM-generated action plans?	Conformal prediction (embodied/ robotics application)	Statistical guarantee on task completion while minimizing human help requests (10–24% less help than baselines without formal assurances); predates #20 by about ten months	Domain-specific (robotics); guarantee is over scenarios, not per episode, and assumes text-grounded environment and faithful human help
34	Conformal Language Modeling (Quach, Fisch, Schuster, Yala, Sohn, Jaakkola, Barzilay)	How can conformal prediction apply to generation, where the output space is unbounded and can only be sampled?	Conformal prediction (foundational, generation-adapted)	Calibrated sampling and rejection give a set that contains an acceptable answer with high probability (i.i.d., admission-function assumptions); set size minimized empirically, plus component-level guarantees	Predates and is more general than later abstention-specific work
35	UProp (Duan, Diffenderfer, Madireddy, Chen, Kailkhura, Xu)	How should an agent's uncertainty at one step account for uncertainty inherited from earlier decisions?	Agentic uncertainty propagation (mutual-information-based)	Outperforms seven single-turn UQ baselines (average or RMS aggregation) by 2.3–11% AUROC on AgentBench-OS and StrategyQA/ HotpotQA; concrete estimator for the propagation problem Oh et al. (#6) later analyze	Relies on Monte-Carlo trajectory sampling (estimates may deviate with few samples; adds latency); partly evaluated on closed-source models

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
36	SAUP (Zhao, Li, Liu, Cheng, Sun, et al.)	Same propagation problem as #35, via a different mechanism	Agentic uncertainty propagation (situational weighting)	Up to ≈0.19 AUROC over best one-step baseline (StrategyQA, LLaMA3 8B); only 0.006–0.016 on GPT-4o HotpotQA/ MMLU; plain RMS/ geometric-mean propagation already improves on one-step baselines	Learned version needs manually annotated data (authors); one-step estimator errors propagate (authors); only length-normalized entropy tested; QA with ReAct, final-answer correctness only
37	Towards Uncertainty-Aware Language Agent / UALA (Han, Buntine, Shareghi)	When should an agent trust its own answer, and when should it use a tool or ask a human?	Action-selection via uncertainty	Vs. always-on ReAct: ChatGPT HotpotQA EM 32.0 to 38.2 with tool calls 2,114 to 403, over 65% fewer tokens; gain shrinks with model size; mixed vs. no-tool baselines (e.g., LLaMA2-70B MMLU 56.7 vs. 64.7 Standard); verbalised confidence worse (HotpotQA, ChatGPT only)	Task-specific uncertainty method and calibration set; threshold from plain-prompt answers may not suit trajectories; QA only, few tools; human help simulated with gold answers

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
38	Perceptions of Linguistic Uncertainty by LMs and Humans (Belém, Kelly, Steyvers, Singh, Smyth)	Do LLMs interpret hedge words (e.g. 'probably') the way humans do, including judging another agent's uncertainty?	Human-perception empirical study	7 of 10 models match population modal answers better than an individual person does, but answers are less varied than people's; for the 6 best models, mean answers are 7.0 points lower for false than true statements (humans: far smaller gap); agreement falls 4.3 points vs. 0.9 for humans on checkable statements	14 English expressions, US-based human reference (94 people), 60+60 statements (400 more as check); truth value is a proxy for model belief; context effects not explored; no explanation of why models acquired the ability
39	"I'm Not Sure, But..." (Kim, Liao, Vorvoreanu, Ballard, Vaughan)	Does an LLM expressing uncertainty in natural language actually change user reliance and trust?	Human-perception empirical study	First-person hedges lowered agreement (74.8% vs. 80.9%) and raised accuracy (72.8% vs. 63.9%) and lowered trust intention; general-perspective hedges moved the same way but not significantly; overreliance reduced, not eliminated	Yes/no medical questions, 8 per participant, U.S. Mechanical Turk workers; fictional system with 50% accuracy and hedges unrelated to correctness; first-person versus general-perspective difference significant only for trust intention

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
40	Revisiting Epistemic Markers in Confidence Estimation (Liu, Zong, Wang, Song)	Does the accuracy attached to the epistemic markers (e.g. 'fairly confident') an LLM uses stay stable across datasets?	Self-verbalized / linguistic marker reliability	7 models, 7 datasets: cross-dataset calibration error 17.7 vs 8.2 in-distribution (numerical confidence 16.6); marker rankings unstable (rank correlation 0.21); stronger models more stable	English only, short closed-form QA, idealized setting; measures the model's own marker accuracy, not human interpretation; no human participants
41	LLMs Know More Than They Show (Orgad, Toker, Gekhman, Reichart, Szpektor, Kotek, Belinkov)	Where in a generated answer is truthfulness encoded, and does a probe trained for one task transfer?	Mechanistic / internal-state probing (supervised, exact answer tokens)	Probe at exact answer tokens: AUC 0.85 vs. 0.75 (minimum token logit) and 0.74 (P(True)) on TriviaQA with Mistral-7b-Instruct; probes generalize mainly between similar tasks and rarely exceed logits alone; error types predictable from internal states	Needs white-box access and labeled data per task; weak cross-task transfer; correctness labels mostly heuristic; refusals excluded; four 7–8B models

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
42	Uncertainty Quantification of LLMs through Multiple Uncertainty Sources (Chen, Liu, Da, Chen, Papalexakis, Wei)	Black-box UQ methods each rely on a single signal and capture one facet of uncertainty	Response-similarity: stacks several sources' similarity matrices into a tensor and scores by Tucker/CP reconstruction error	Higher AUROC and AUARC than the best single-source baseline in all 30 cells of the main table (e.g., CoQA, Llama2-13B AUROC: 0.7802 vs. 0.7566)	Needs auxiliary LLMs and an NLI model (added cost); a badly wrong source can degrade the combined score; some margins very small; no variance or significance tests
43	Judge, Retrieve, or Abstain (Badshah, Emami, Sajjad)	LLM judges of factual QA err silently, and uncertainty thresholds give no control over the error rate of accepted verdicts	Risk-controlled routing: predictive entropy, retrieve and re-judge, else abstain; Clopper–Pearson bound on the false discovery rate	Bound held in all 32 candidate–judge–dataset combinations; at target 0.20, retrieval raised coverage 0.40 to 0.85 (HotpotQA, Llama-8B judge) and 0.07 to 0.82 (NQ-Open, Qwen-8B judge)	Factual QA only; token log-probabilities needed; recalibration when task, candidate, or judge changes; data-dependent threshold search (main results uncorrected); labels mostly from an LLM evaluator; human review not implemented

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Table S1 – continued

#	Paper	Problem	UQ Technique	Key Results	Limitations
44	Not All Uncertainty Is Equal (Villavicencio, Pan, Wang)	Does the granularity of displayed uncertainty change how people verify LLM answers?	Human-perception empirical study: output-, step-, and word-level uncertainty displays	N=192 in four conditions; word-level display raised agreement (0.730 vs. 0.674); step-level display cut link clicks (0.318 vs. 0.557); no significant accuracy differences (0.617–0.657)	Uncalibrated, overconfident estimates; deliberately hard medical questions; crowdsourced U.S., English-speaking, highly educated sample
45	Anthropomimetic Uncertainty (Ulmer, Lorson, Titov, Hardmeier)	Verbalized uncertainty in LLMs ignores how humans communicate uncertainty	Argument paper combining a review of 53 papers with experiments on six commercial LLMs	61% of reviewed methods use numerical expressions and 78% prompting; verbalization varies widely with model, speaker–listener relationship, and subject; calibration errors up to 44%	English only; correctness judged by an LLM plus heuristics; no participant data of its own; no access to pretraining data

Table S2. Per-paper coding used for the counts in Section 5.1 of the article. Numbers follow Table S1.

#	Paper	Year	Setting	Human	Task type	Primary metric	Guar.
1	Kadavath et al.	2022	single	–	mixed	calibration	–
2	Manakul et al. (SelfCheck-GPT)	2023	single	–	open-ended	discrimination	–
3	Kuhn et al.	2023	single	–	open-ended	discrimination	–
4	He et al.	2026	single	–	n/a	n/a	–
5	Zhang et al. (SELAUR)	2026	agentic	–	non-QA	other	–
6	Oh et al.	2026	agentic	–	non-QA	discrimination	–
7	Geng et al.	2024	single	–	n/a	n/a	–
8	Donkers and Ziegler	2026	single	yes	non-QA	other	–
9	Shorinwa et al.	2025	single	–	n/a	n/a	–
10	Shrivastava et al.	2025	single	–	fixed-choice	discrimination	–
11	Zhang et al. (SAC3)	2023	single	–	mixed	discrimination	–
12	Nikitin et al.	2024	single	–	open-ended	discrimination	–
13	Xiong et al.	2024	single	–	mixed	discrimination	–
14	Duan et al. (SAR)	2024	single	–	open-ended	discrimination	–
15	Vashurin et al.	2025	single	–	mixed	discrimination	–

#	Paper	Year	Setting	Human	Task type	Primary metric	Guar.
16	Kirchhof et al.	2025	single	–	n/a	n/a	–
17	Fadeeva et al.	2024	single	–	open-ended	discrimination	–
18	Lin et al. (Generating with Confidence)	2024	single	–	open-ended	discrimination	–
19	Quevedo et al.	2025	single	–	open-ended	other	–
20	Abbasi-Yadkori et al.	2024	single	–	open-ended	coverage/risk	yes
21	Tayebati et al.	2025	single	–	fixed-choice	coverage/risk	yes
22	Pesaranghader and Li	2026	single	–	n/a	n/a	–
23	Rubashevskii et al.	2026	single	–	mixed	coverage/risk	yes
24	Devic et al.	2025	single	–	n/a	n/a	–
25	Lin et al. (Teaching Models)	2022	single	–	open-ended	calibration	–
26	Azaria and Mitchell	2023	single	–	mixed	other	–
27	Burns et al.	2023	single	–	fixed-choice	other	–
28	Farquhar et al.	2024	single	–	open-ended	discrimination	–
29	Kossen et al.	2024	single	–	open-ended	discrimination	–
30	Mündler et al.	2024	single	–	open-ended	other	–
31	Ye et al.	2024	single	–	mixed	coverage/risk	yes
32	Li et al. (ITI)	2023	single	–	mixed	other	–
33	Ren et al. (KnowNo)	2023	agentic	–	non-QA	coverage/risk	yes
34	Quach et al.	2024	single	–	open-ended	coverage/risk	yes
35	Duan et al. (UProp)	2025	agentic	–	non-QA	discrimination	–
36	Zhao et al. (SAUP)	2025	agentic	–	non-QA	discrimination	–
37	Han et al. (UALA)	2024	agentic	–	non-QA	other	–
38	Belém et al.	2024	single	yes	non-QA	other	–
39	Kim et al.	2024	single	yes	non-QA	other	–
40	Liu et al.	2025	single	–	fixed-choice	calibration	–
41	Orgad et al.	2025	single	–	open-ended	discrimination	–
42	Chen et al.	2026	single	–	open-ended	discrimination	–
43	Badshah et al.	2026	single	–	open-ended	coverage/risk	yes
44	Villavicencio et al.	2026	single	yes	non-QA	other	–
45	Ulmer et al.	2026	single	–	open-ended	calibration	–

Notes on judgment calls for papers 43–45. Badshah et al. (43) are coded open-ended because the judged answers are generated answers to open-domain questions, although the paper does not describe them as free-form and the judge itself outputs a binary verdict. Villavicencio et al. (44) are coded non-QA because the task is human verification behavior, as for Belém et al. and Kim et al. Ulmer et al. (45) are coded calibration because the paper names no single headline result and its introduction summarizes its experiments by calibration errors of up to 44%, and open-ended because the only answer format it states is free text (TriviaQA); the answer format of its MMLU experiment is not stated.

Table S3. Update-search log (searched 14 October 2026). Sources: arxiv.org, aclanthology.org, dl.acm.org, openreview.net. Up to ten results per query; screened by title and snippet only; four of the papers it found were later added to the corpus (Section 2 of the article, fourth stage).

#	**Query string**
1	uncertainty quantification large language models 2026
2	hallucination detection semantic entropy large language models 2026
3	confidence calibration verbalized confidence large language models 2026
4	conformal prediction abstention large language models hallucination 2026
5	uncertainty-aware mitigation hallucination LLM abstain retrieve clarify 2026
6	uncertainty quantification LLM agents multi-turn tool use 2026
7	LLM uncertainty expression user trust reliance human study 2026
8	survey uncertainty estimation calibration large language models
9	token probability entropy uncertainty estimation LLM free-form generation
10	internal states probing truthfulness hallucination detection LLM hidden states
11	black-box uncertainty estimation consistency sampling LLM
12	selective prediction abstention language models risk control guarantee
13	uncertainty-guided retrieval when to retrieve LLM confidence
14	reinforcement learning calibration reward confidence LLM uncertainty training
15	clarification questions uncertainty ambiguity LLM agent ask user
16	human-centered uncertainty communication large language models user study
17	epistemic markers hedging linguistic uncertainty LLM human perception
18	agent uncertainty propagation trajectory-level confidence calibration
19	benchmark evaluating uncertainty quantification methods LLM

Table S4. Dispositions of the 126 distinct records from the update search (searched 14 October 2026). Dispositions and pillar labels (Det. = Detection, Mit. = Mitigation, HC&A = Human-Centered and Agentic) were assigned from titles and search snippets only and are provisional; titles are abbreviated and identifiers are as returned by the search.

Record (abbreviated title)	Identifier	Disposition	Pillar
INSIDE	arXiv 2402.03744	Eligible (new)	Det.
Benchmarking Uncertainty with MCQ	arXiv 2406.12784	Eligible (new)	Det.
LLM Internal States Reveal Hallucination Risk	arXiv 2407.03282	Eligible (new)	Det.
Universal Truthfulness Hyperplane	arXiv 2407.08582	Eligible (new)	Det.
LLMs Know More Than They Show	arXiv 2410.02707	Added to corpus after search	Det.
DiverseAgentEntropy	arXiv 2412.09572	Eligible (new)	Det.
UQ for Language Models: Scorers Suite	arXiv 2504.19254	Eligible (new)	Det.
Token-Level UQ for LLM Reasoning	arXiv 2505.11737	Eligible (new)	Det.
Are the Hidden States Hiding Something?	arXiv 2505.16520	Eligible (new)	Det.
Revisiting Uncertainty Estimation and Calibration	arXiv 2505.23854	Eligible (new)	Det.
Consistency Hypothesis in UQ for LLMs	arXiv 2506.21849	Eligible (new)	Det.
Rehearse	arXiv 2508.14390	Eligible (new)	Det.
Calibrating Verbalized Confidence with Distractors	arXiv 2509.25532	Eligible (new)	Det.
Do LLMs Really Know What They Dont Know?	arXiv 2510.09033	Eligible (new)	Det.
ADVICE	arXiv 2510.10913	Eligible (new)	Det.
Systematic Evaluation of UE Methods	arXiv 2510.20460	Eligible (new)	Det.
Probabilities Are All You Need	arXiv 2511.07694	Eligible (new)	Det.
Mathematical Analysis of Hallucination Dynamics	arXiv 2511.15005	Eligible (new)	Det.
Fact-Checking via Probabilistic Certainty and Consistency	arXiv 2601.02574	Eligible (new)	Det.
Semantic UQ of Hallucinations: Quantum Tensor Network	arXiv 2601.20026	Eligible (new)	Det.
Benchmarking Calibration in Long-Form QA	arXiv 2602.00279	Eligible (new)	Det.
Benchmark of Uncertainty Metrics for Automatic Grading	arXiv 2602.16039	Eligible (new)	Det.
UE Scales with Sampling in Reasoning Models	arXiv 2603.19118	Eligible (new)	Det.
Adaptive Bayesian Estimation of Semantic Entropy	arXiv 2603.22812	Eligible (new)	Det.
Closing the Confidence-Faithfulness Gap	arXiv 2603.25052	Eligible (new)	Det.
UQ for LLM Function-Calling	arXiv 2604.22985	Eligible (new)	Det.
ORCE	arXiv 2605.12446	Eligible (new)	Det.
UQ for Large Language Diffusion Models	arXiv 2605.14570	Eligible (new)	Det.
MultiHaluDet	arXiv 2605.24919	Eligible (new)	Det.
Entropy Distribution as Fingerprint for Hallucinations	arXiv 2605.28264	Eligible (new)	Det.
BEACON	arXiv 2606.07528	Eligible (new)	Det.
Systematic Evaluation of Black-Box UE	arXiv 2606.19868	Eligible (new)	Det.
The Origins of Stochasticity: UQ for LLMs	arXiv 2606.22792	Eligible (new)	Det.
Hallucination Detection using Diversion Decoding	arXiv 2607.10476	Eligible (new)	Det.
ConfidenceBench	arXiv 2607.20526	Eligible (new)	Det.
Propagating Uncertainty Across Reasoning and Retrieval	arXiv 2607.25600	Eligible (new)	Det.
Claim-Level Confidence Calibration	arXiv 2608.22483	Eligible (new)	Det.
Truthfulness Signal in Code-Mixing	arXiv 2609.22138	Eligible (new)	Det.
Cross-Model Span-Level Hallucination Detection	arXiv 2610.02066	Eligible (new)	Det.
UQ of LLMs through Multiple Uncertainty Sources	ACL Findings 2026, 2026.findings-acl.535	Added to corpus after search	Det.
Beyond Semantic Entropy	ACL Findings 2025, 2025.findings-acl.234	Eligible (new)	Det.
Can LLMs Learn Uncertainty on Their Own?	EMNLP 2024, 2024.emnlp-main.1205	Eligible (new)	Det.
UQ for In-Context Learning of LLMs	NAACL 2024, 2024.naacl-long.184	Eligible (new)	Det.

Record (abbreviated title)	Identifier	Disposition	Pillar
Consistency-based Black-box UQ for Text-to-SQL	OpenReview ofiZbAmrZh	Eligible (new)	Det.
Humans Overrely on Overconfident LMs	arXiv 2507.06306	Eligible (new)	HC&A
What Verbalized Uncertainty in Language Models is Missing	arXiv 2507.10587 (TACL 2026 version cited)	Added to corpus after search	HC&A
Can LLMs Express Uncertainty Like Human?	arXiv 2509.24202	Eligible (new)	HC&A
Agentic Uncertainty Quantification	arXiv 2601.15703	Eligible (new)	HC&A
Agentic Confidence Calibration	arXiv 2601.15778	Eligible (new)	HC&A
LLMs Should Express Uncertainty Explicitly	arXiv 2604.05306	Eligible (new)	HC&A
UQ of LLM Multi-Agent Systems (Tensor Decomposition)	arXiv 2604.08708	Eligible (new)	HC&A
Proper Scoring Rules for Agentic UQ	arXiv 2605.24756	Eligible (new)	HC&A
Knowing but Not Showing	arXiv 2605.25284	Eligible (new)	HC&A
Not All Uncertainty Is Equal	arXiv 2605.28571 (FAccT 2026 version cited)	Added to corpus after search	HC&A
Linguistic Uncertainty Markers and Intrinsic Confidence	arXiv 2605.28778	Eligible (new)	HC&A
Confidence Laundering in Agent Systems	arXiv 2606.20662	Eligible (new)	HC&A
U-Lens	arXiv 2607.10604	Eligible (new)	HC&A
Trajectory-Adapted UQ for LLM Agents	arXiv 2608.11552	Eligible (new)	HC&A
Relational Uncertainty Propagation for Agents	arXiv 2608.16002	Eligible (new)	HC&A
PropUQ-MAS	arXiv 2608.22130	Eligible (new)	HC&A
UQ for LLM Agents: Taxonomy and Empirical Study	arXiv 2609.07395	Eligible (new)	HC&A
Do Agents Know When They Succeed?	arXiv 2609.09448	Eligible (new)	HC&A
GRUET	arXiv 2609.24831	Eligible (new)	HC&A
Clarify the User or Verify the World?	arXiv 2609.32255	Eligible (new)	HC&A
Structured Uncertainty guided Clarification	ACL Findings 2026, 2026.findings-acl.2028	Eligible (new)	HC&A
Impact of LMs Reluctance to Express Uncertainty	ACL 2024, 2024.acl-long.198	Eligible (new)	HC&A
Confronting verbalized uncertainty (IJHCS)	IJHCS vol. 197, 103455	Eligible (new)	HC&A
Selective Generation for Controllable Language Models	arXiv 2307.09254	Eligible (new)	Mit.
When Do LLMs Need Retrieval Augmentation?	arXiv 2402.11457	Eligible (new)	Mit.
Adaptive Retrieval without Self-Knowledge?	arXiv 2501.12835	Eligible (new)	Mit.
Rewarding Doubt	arXiv 2503.02623	Eligible (new)	Mit.
Do not Abstain! Identify and Solve the Uncertainty	arXiv 2506.00780	Eligible (new)	Mit.
Beyond Binary Rewards	arXiv 2507.16806	Eligible (new)	Mit.
TECP: Token-Entropy Conformal Prediction	arXiv 2509.00461	Eligible (new)	Mit.
Selective Risk Certification via Information-Lift	arXiv 2509.12527	Eligible (new)	Mit.
Domain-Shift-Aware Conformal Prediction for LLMs	arXiv 2510.05566	Eligible (new)	Mit.
Behaviorally Calibrated RL	arXiv 2512.19920	Eligible (new)	Mit.
Calibration Aware RL	arXiv 2601.13284	Eligible (new)	Mit.
Black-Box Reliability Certification for Agents	arXiv 2602.21368	Eligible (new)	Mit.
I-CALM	arXiv 2604.03904	Eligible (new)	Mit.
Geometry-Calibrated Conformal Abstention	arXiv 2604.27914	Eligible (new)	Mit.
Calibrating LLMs with Semantic-level Reward	arXiv 2605.15588	Eligible (new)	Mit.
Agentic Tool-Calling via Uncertainty-Aligned RL	arXiv 2606.06976	Eligible (new)	Mit.
RL with Metacognitive Feedback	arXiv 2606.32032	Eligible (new)	Mit.
Uncertainty-Aware Abstention with Provable Guarantees	arXiv 2607.04430	Eligible (new)	Mit.
Hierarchical Group-Conditional Conformal Risk Control	arXiv 2607.24562	Eligible (new)	Mit.
Judge, Retrieve, or Abstain	arXiv 2608.17994	Added to corpus after search	Mit.
Can Language Models Learn When to Abstain?	arXiv 2610.13821	Eligible (new)	Mit.
Kuhn semantic uncertainty	arXiv 2302.09664	Already in corpus	–
Geng survey	arXiv 2311.08298	Already in corpus	–
Ye benchmarking via UQ	arXiv 2401.12794	Already in corpus	–
Mitigating Hallucinations via Conformal Abstention	arXiv 2405.01563	Already in corpus	–

Record (abbreviated title)	Identifier	Disposition	Pillar
Learning Conformal Abstention Policies (also OpenReview version)	arXiv 2502.06884	Already in corpus	–
Kirchhof position paper	arXiv 2505.22655	Already in corpus	–
Hallucination Detection and Mitigation in LLMs	arXiv 2601.09929	Already in corpus	–
UQ in LLM Agents: Foundations (Oh et al.)	arXiv 2602.05073	Already in corpus	–
Shorinwa survey	doi 10.1145/3744238	Already in corpus	–
Kim et al. I am Not Sure, But	arXiv 2405.00623	Already in corpus	–
Donkers polarized discourse	doi 10.1145/3774935.3806174	Already in corpus	–
LM-Polygraph	arXiv 2406.15627	Already in corpus	–
UQ and Confidence Calibration: A Survey	arXiv 2503.15850	Excluded: survey/review	–
Disambiguation in Conv. QA: Survey	arXiv 2505.12543	Excluded: survey/review	–
Theoretical Foundations and Mitigation of Hallucination	arXiv 2507.22915	Excluded: survey/review	–
LLM Hallucination: A Comprehensive Survey	arXiv 2510.06265	Excluded: survey/review	–
UQ for Hallucination Detection: Foundations, Methodology	arXiv 2510.12040	Excluded: survey/review	–
Uncertainty-Aware Decision Making in MLLMs: Survey	arXiv 2608.17084	Excluded: survey/review	–
Survey of UE Methods on LLMs	ACL Findings 2025, 2025.findings-acl.1101	Excluded: survey/review	–
UQ for Large Language Models (book chapter)	doi 10.1007/978-3-032-21321-1_7	Excluded: survey/review	–
Abstention Knobs for Video QA	arXiv 2601.00138	Excluded: multimodal	–
UQ for Multimodal LLMs (semantic volume)	arXiv 2602.24195	Excluded: multimodal	–
Segmentation-Based Attention Entropy (vision-language)	arXiv 2603.16558	Excluded: multimodal	–
Look Again Before You Abstain (vision-language)	arXiv 2606.16667	Excluded: multimodal	–
Training-Free UE in Multimodal LLMs	arXiv 2609.22206	Excluded: multimodal	–
VL-Calibration (vision-language)	ACL 2026, 2026.acl-long.2074	Excluded: multimodal	–
Learning to Ask (unclear instruction)	arXiv 2409.00557	Excluded: off topic	–
ClarQ-LLM	arXiv 2409.06097	Excluded: off topic	–
Quantifying Uncertainty in LLM Benchmark Scores	arXiv 2410.03492	Excluded: off topic	–
Curiosity by Design	arXiv 2507.21285	Excluded: off topic	–
Prospect Theory for LLMs	arXiv 2508.08992	Excluded: off topic	–
Unidentified reward-pipeline record	arXiv 2603.06317	Excluded: off topic	–
Seeing the Reasoning: Rationales and Trust	arXiv 2603.07306	Excluded: off topic	–
Conformal Selective Prediction with General Risk Control	arXiv 2603.24704	Excluded: off topic	–
Label Effects: Trust, LLM-as-a-Judge	arXiv 2604.05593	Excluded: off topic	–
ASPI prompt injection	arXiv 2605.17324	Excluded: off topic	–
LLM or Human? Trust in Research Summaries	doi 10.1145/3772318.3793386	Excluded: off topic	–
Unidentified OpenReview PDF	OpenReview PDF (title not shown)	Excluded: off topic	–

Table S5. Cost and deployment constraints by method family, synthesized from each family's own reported requirements.

Method Family	Model Access Required	Model Calls per Query	Auxiliary Model / Training Data	Calibration Data / Human Involvement	Latency Profile
Token-level (Section 4.2 of the article)	White- or grey-box (logits)	1 (single forward pass)	None beyond the base model	None	Fastest; no extra sampling
Semantic-similarity (Section 4.2 of the article)	Mostly black-box (sampled responses only); grey-box where token probabilities are used (e.g., Semantic Entropy [2])	Multiple (typically 5–20 sampled generations, e.g., ~10 for Semantic Entropy [2], ~20 for SelfCheckGPT [3])	Typically an NLI or entailment model	None (unsupervised at inference)	Slow relative to token-level; scales with sample count
Self-verbalized / introspective (Section 4.2.6 of the article)	Black-box (verbalized) or white-box (P(IK)-style probes)	Low; typically 1–2 extra calls (P(True)-style), more if aggregated over samples	A trained internal-state probe for P(IK)-style variants only	Labeled examples to train the probe (white-box variants only); none for verbalized variants	Fast for verbalized variants; probe training is a one-time offline cost
Mechanistic / internal-state (Section 4.2 of the article)	White-box (hidden activations)	1 (probe/intervention reuses the existing forward pass)	A trained probe or intervention direction	Labeled or unsupervised training data to fit the probe/direction	Fast at inference; upfront probe-training cost
Conformal prediction (Section 4.3 of the article)	Varies by base estimator (white- or black-box)	Same as the underlying base estimator; none extra for calibration itself	None beyond the base estimator	A held-out calibration set (Yadkori et al. [1] tried 10 to 3,200 examples and saw near-total abstention at the smallest sizes in some settings); a human in the loop for the embodied clarification variant [4] only	Fast once calibrated; calibration is a one-time or periodically repeated offline cost
Training-time / RL-based (Section 4.3 of the article)	White-box (training-time logits)	N/A extra at inference; cost is paid during training	Full RL training infrastructure and reward-shaping design	None at inference; substantial labeled/simulated training data during training	No added inference latency; substantial upfront training cost

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